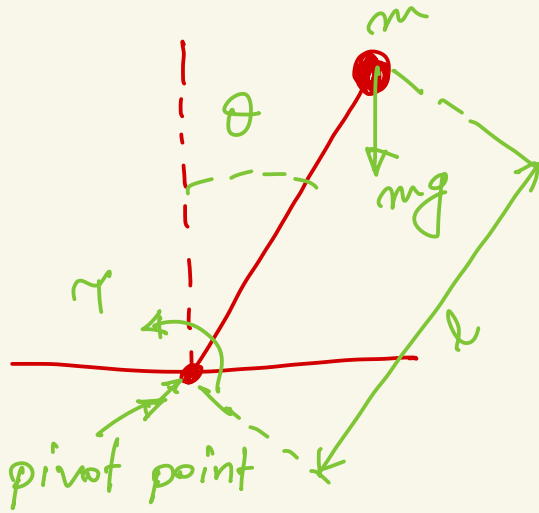




Inverted pendulum on a hinge:

Stability

Control



Balance of moment across the pivot point gives the below equation

$$ml^2\ddot{\theta} - mgl\sin\theta = \tau$$

$$\Rightarrow \ddot{\theta} - \frac{g}{l}\sin\theta = \frac{\tau}{ml^2}$$

For small angle approximation

$$\sin \theta \approx \theta$$

$$\Rightarrow \ddot{\theta} - \frac{g}{l} \theta = \gamma \cdot \frac{1}{ml^2}$$

Take Laplace transform:

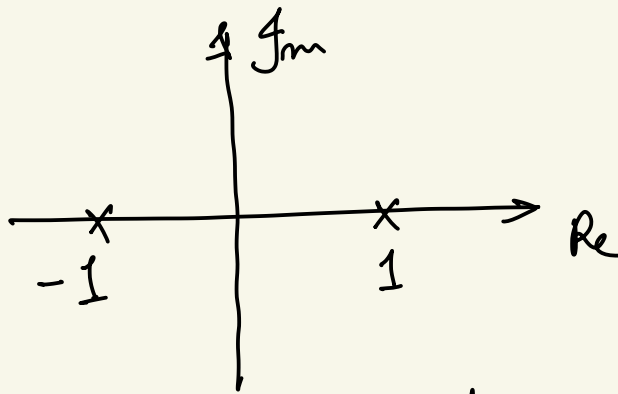
$$\Rightarrow s^2 \theta(s) - \frac{g}{l} \theta(s) = \gamma(s) \cdot \frac{1}{ml^2}$$

$$\Rightarrow \frac{\theta(s)}{\gamma(s)} = \frac{1}{s^2 - \frac{g}{l}} \cdot \frac{1}{ml^2}$$

Assume $\frac{g}{l} = 1$, $ml^2 = 1$

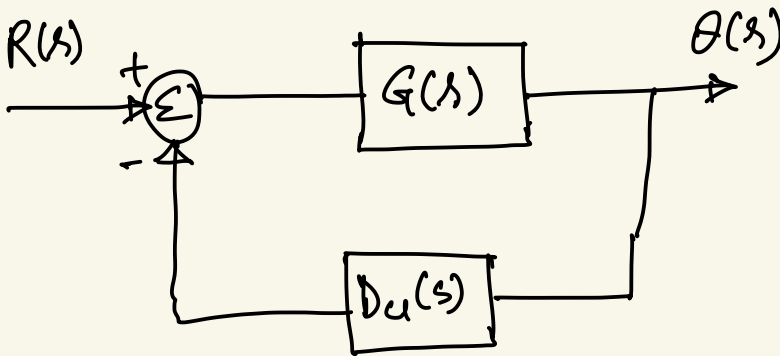
$$\Rightarrow \frac{\theta(s)}{\gamma(s)} = G(s) = \frac{1}{s^2 - 1}$$

$$G(s) = \frac{1}{(s+1)(s-1)}$$



System is unstable.

Design a controller $D_c(s)$ to make the system stable.



From previous lecture, the closed loop transfer function $T_{cl}(s)$

$$T_c(s) = \frac{\Theta(s)}{R(s)} = \frac{G(s)}{1 + G(s)D_c(s)}$$

Assume, a dynamic compensator controller $D_c(s) = K \frac{s + \gamma}{s + \delta}$

There are 3 parameters that can be tuned. K, γ, δ .

In order to have the system stable, the denominator of the closed loop transfer function is set equal to zero and poles are investigated.

$$1 + G(s) D_c(s) = 0$$

$$\Rightarrow 1 + \frac{1}{s-1} \cdot k \frac{s+\gamma}{s+\delta} = 0$$

$$\text{set, } \gamma = 1$$

$$\Rightarrow 1 + \frac{1}{(s-1)\cancel{(s+1)}} \cdot k \frac{\cancel{(s+1)}}{(s+\delta)} = 0$$

$$\Rightarrow 1 + \frac{k}{(s-1)(s+\delta)} = 0$$

$$\Rightarrow (s-1)(s+\delta) + k = 0$$

$$\Rightarrow s^2 + (\delta-1)s + (k-\delta) = 0$$

$$s_{1,2} = \frac{-(\delta-1) \pm \sqrt{(\delta-1)^2 - 4 \cdot 1 \cdot (k-\delta)}}{2}$$

Assume, $K = 3$, $\delta = 2$

$$s_{1,2} = \frac{-(2-1) \pm \sqrt{(2-1)^2 - 4 \cdot (3-2)}}{2}$$

$$= \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= \frac{-1 \pm 3j}{2}$$

Now the system is stable.

The recommended controller $D_{cl}(s)$ is :

$$D_{cl}(s) = 3 \frac{s+1}{s+2}$$

$$\frac{\Theta(s)}{R(s)} = \frac{G(s)}{1 + G(s)D_{cl}(s)}$$

$$= \frac{\frac{1}{s^2-1}}{1 + \frac{1}{s^2-1} \cdot 3 \frac{(s+1)}{(s+2)}}$$

$$T_{cl}(s) = \frac{\Theta(s)}{R(s)} = \frac{(s+2)}{s^3 + 2s^2 + 2s + 1}$$

The resulting closed-loop transfer function is stable.

